THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 2: **Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of the zeroes of $x^2 + 5x + 6$ is: 1

(a) -5 **(b)** -6 **(c)** -4 **(d)** -3

2. The product of the zeroes of $x^2 + 5x + 6$ is: 1

(a) 7 **(b)** 6 **(c)** 5 **(d)** 8

3. A polynomial of degree 5 has at most how many zeroes? 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** infinitely many

4. Is 3 a zero of $2x^2 - 3x + 1$? 1

(a) Yes **(b)** Cannot say **(c)** No **(d)** Only if $k > 0$

5. If $p(x) = x^2 + 4x + 4$, then $p(1)$ is: 1

(a) 10 **(b)** 11 **(c)** 9 **(d)** 8

6. The zero of the linear polynomial $7x - 21$ is: 1

(a) -21 **(b)** 3 **(c)** 21 **(d)** -3

7. The sum of the zeroes of $x^2 - 8x + 7$ is: 1

(a) 9 **(b)** 8 **(c)** 10 **(d)** 7

8. The product of the zeroes of $2x^2 - 16x + 14$ is: 1

(a) 6 **(b)** 9 **(c)** 7 **(d)** 8

9. Is -4 a zero of $x^2 - 16$? 1

(a) Yes **(b)** No **(c)** Only if $k > 0$ **(d)** Cannot say

10. If $p(x) = 2x^2 - 7x + 3$, then $p(5)$ is: 1

(a) 18 **(b)** 17 **(c)** 20 **(d)** 19

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the zeroes of $x^2 - 7x + 6$. 2

12. Find the value of $x^2 - x + 1$ at $x = 5$. 2

13. Find the zero of the polynomial $4x + 6$. 2

14. Find the zeroes of $x^2 + 9x + 14$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the zeroes of $x^2 + 7x + 12$ by splitting the middle term and verify the sum of zeroes. 3

16. Find a quadratic polynomial whose sum and product of zeroes are 4 and -21 respectively. 3

17. Verify that 2 is a zero of $x^3 - 10x^2 + 31x - 30$. 3

OR

Find the zeroes of $x^2 - 10x + 9$ by splitting the middle term and verify the sum of zeroes. 3

18. Find a quadratic polynomial whose sum and product of zeroes are -8 and 15 respectively. 3

OR

Verify that -2 is a zero of $x^3 - 5x^2 - 8x + 12$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the zeroes of $2x^2 - 7x + 3$ and verify the relationship between the zeroes and the coefficients. 5

OR

Verify the relationship between zeroes and coefficients for the cubic $x^3 - 12x^2 + 44x - 48$ whose zeroes are $2, 4, 6$. 5

20. Find the zeroes of $3x^2 + x - 2$ and verify the relationship between the zeroes and the coefficients. 5

OR

Verify the relationship between zeroes and coefficients for the cubic $x^3 - 7x - 6$ whose zeroes are $-2, -1, 3$. 5

END OF QUESTION PAPER — Total: 40 Marks

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